Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Scientist with 3 years of industry experience in asset management (HDFC Asset Management), plus an MSc in Data Science (University of Bristol). Analysed fund, credit-rating and customer data with **Python** and **SQL**, applied **Market Basket Analysis (Apriori)** for the marketing team, built a pipeline that parses 300+ daily rating-agency emails and saves 60 analyst hours a month, and moved financial analysts from Excel to **Power BI**, improving efficiency by 10 hours a week. Worked directly with Investment, Compliance and Client Service from requirements to release, presenting results to non-technical users, and acted as de facto team lead for up to 4 interns. The co-built AstraZeneca dissertation project ranks cancer cell lines with a multi-evidence scoring algorithm refined with scientist feedback; 3rd place at a Lloyds Bank-sponsored fraud detection datathon.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Analytics, Modelling & Reporting

- Analyzed customer data with **Python Pandas**, used in virtually every project, and identified customer buying patterns using **Market Basket Analysis** by implementing the **Apriori Algorithm**, handing the results to the marketing team as insights for them to act on.
- Built a **Python** pipeline that parses over **300 daily emails** from credit rating agencies (ICRA, India Ratings, CRISIL) into structured data, uses **fuzzywuzzy** to fuzzy-match company names against those analysts track and sends a daily consolidated email of previous and changed ratings, **saving 60 manual analyst hours a month**; first built on **AWS Textract**, then moved to **Azure OpenAI** for better accuracy when an agency began sending emails as images.
- Developed **Power BI** dashboards for the investment team's fortnightly credit report analysis, migrating financial analysts from **Excel to Power BI** and improving efficiency by **10 hours a week**; recreated the dashboards in the **ReactJS** frontend so the nationwide Client Service team could use the data without Power BI licensing costs.

Data Foundations for Analytics & AI

- Designed the **Normalized SQL Schema** (PostgreSQL) for the internal analytics platform to the analysts' requirements, with **Python** scripts ingesting the licensed **Morningstar API** for every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly (fixed income) and monthly (equity), and integrated with the legacy analyst data previously held in Excel; wrote **Optimized SQL** queries for data preprocessing and **SQL Functions** and **Procedures** for efficient data insertion.
- Built the retrieval and tooling layer for **GenAI applications** used by Compliance, Client Service and Investment: **RAG** with **LlamaIndex** and **Azure OpenAI** embeddings over a separate **pgvector** database per department, and tool-calling agents that run **Python** functions and **SQL** queries over the fund data to compare funds and fund performance and generate PDF comparison reports on request.
- Deployed the platform and applications on **AWS** (EC2, S3, ECR, Lambda) as **Docker** containers behind an **Nginx** reverse proxy through a one-click **Jenkins** and **GitLab** CI/CD pipeline; implemented 3-tier **RBAC** with **LDAP**

authentication and network restrictions with the security team, the Nginx proxy being one of the fixes that took the applications through the security team's **VAPT**.

Stakeholders & Leadership

- Worked directly with **Client Service, Compliance and Investment** from requirements to release, building to the analysts' requirements and presenting to non-technical users, with Compliance reviewing every application; built the first systems as the only engineer alongside my manager, later acted as **de facto team lead** for up to 4 interns; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; personal focus on the data preprocessing, scoring algorithm and RAG assistant. Preprocessing pipeline joins 17 raw and augmented inputs (DepMap, GEO, HPA) into 14 clean tables with identity resolution, duplicate and missing-value handling, scaling (log2 TPM, z-scores) and integrity checks; scoring algorithm ranks cancer cell lines by target inclusion and exclusion genes across six evidence layers using weighted desirability scores, correlation weighting and a calibration step, with confidence tiers; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project of 5; multi-phase study comparing CNN, ResNet and ViT encoders with LSTM and attention decoders for generating image descriptions on three synthetic datasets, evaluated over five seeds with SBERT cosine, BLEU and ROUGE-L and benchmarked against pre-trained vision-language models.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Swindon Borough Council Fuel-Poverty Hack:** A household and neighborhood-based fuel poverty data analysis tool. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle **credit card fraud detection** problem.)
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git

SKILLS

- **Tools:** Power BI, Jupyter, AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, CloudFormation templates), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab (CI/CD), Nginx, Linux, LDAP, Git, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, fuzzywuzzy, FastAPI), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (schema design, SQL functions and procedures, pgvector), SQLite